

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
7	Indicative content A1 Partial pressure of oxygen increases percentage saturation also increases A2 (Hb has) high <u>affinity</u> for O₂ at {high pO₂ / lungs} A3 {Associates / loads} with O₂ (in the lungs / high pO₂) A4 (Hb has) low <u>affinity</u> for O₂ {at low pO₂ / tissues} A5 {dissociates / unloads} with O₂ (in the tissues / low pO₂) A6 sigmoid shape / co-operative binding / steep part of the curve – small drop in pO₂ relatively large % drop in saturation / OWTTE <u>Naked mole rat haemoglobin:</u> B1 Lives in a low pO₂ environment B2 due to {poorly ventilated / underground} burrows / large number of individuals respiring B3 Has <u>Hb</u> with a higher <u>affinity</u> for O₂ B4 <u>Hb</u> can become {fully / higher %} saturated with O₂ B5 {At a lower pO₂ / at same pO₂} as human Hb <u>Hummingbird haemoglobin:</u> C1 Smaller birds so have a larger surface area: volume so lose heat at a faster rate / Flight muscles 30% of their body weight / beat their wings up to 200 times per second C2 have a higher rate of respiration C3 this produces more CO₂ / increase pCO₂ C4 Hb with a lower affinity for O₂ C5 so {dissociates / unloads} more O₂ to the tissues / (more) readily releases O₂	3	3	3	9	0	0